

# PyGameMaker — Tutorial 2: Your First Game — Catch the Star

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## How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Your First Game: Catch the Star** (5 pages, about 15-20 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase before moving on. If something does not match, look at **Stuck?** before you call your teacher.

## What You Will Make

A complete little game: you move a spaceship left and right along the bottom of the screen, stars fall from the sky, and every star you catch gives you 10 points.

## Phase 1: A Moving Player

- ☐ 1. Make a new project called CatchTheStar (**File > New Project**).
- ☐ 2. Create a sprite spr\_player (32×32) and draw a spaceship, a person or a basket. Save with **Ctrl+S**.
- ☐ 3. Create an object obj\_player and give it the sprite spr\_player.
- ☐ 4. Create a room room\_game and place obj\_player near the bottom centre.
- ☐ 5. In obj\_player, open the **Blockly** tab and add three events: **Keyboard: Left Arrow (held)** with \*Set horizontal speed to -5\*, **Keyboard: Right Arrow (held)** with \*Set horizontal speed to 5\*, and **Keyboard: No key** with \*Stop movement\*.

**You should see:** press **F5**. The arrow keys move your player left and right, and it stops when you let go. It can leave the screen — that is expected for now.

## Phase 2: Falling Stars

- ☐ 6. Create a sprite spr\_star (32×32) and an object obj\_star that uses it.
- ☐ 7. In obj\_star, add a **Create** event with \*Set vertical speed to 3\*.
- ☐ 8. Create an object obj\_spawner with **no sprite**. Its **Create** event does \*Set Alarm 0 to 60\*. Its **Alarm 0** event does \*Create instance of obj\_star at x: random, y: 0\* and then \*Set Alarm 0 to 60\* again.
- ☐ 9. Place obj\_spawner anywhere in room\_game.

**You should see:** press **F5**. About one star per second appears at the top, in a different place each time, and falls. Stars pass straight through your player — that is expected for now.

## Phase 3: Catching and Scoring

- ☐ 10. In obj\_player, add **Collision with obj\_star**: \*Add to score 10\*, then \*Destroy other instance\*.
- ☐ 11. Create obj\_game\_controller (no sprite). Its **Create** event does \*Set score to 0\*; its **Draw** event does \*Draw score at x: 10, y: 10\*.
- ☐ 12. In obj\_star, add an **Outside Room** event with \*Destroy this instance\*.

- ☐ **13.** Place `obj_game_controller` anywhere in the room.

**You should see:** press **F5**. Catch a star: it disappears and the score at the top left goes up by 10. Stars you miss disappear at the bottom.

## Phase 4: Finishing Touches

- ☐ **14.** In `obj_player`, add a **Step** event that keeps the player on the screen: if `x` is less than 0, set `x` to 0; if `x` is greater than the room width minus the sprite width, set `x` to that value.
- ☐ **15.** Give `room_game` a background colour you like.

**You should see:** the player stops at both edges of the window, and your finished game is ready to play.

## Stuck?

| What is wrong                               | Most likely cause                                                                       | What to do                                                                                                     |
|---------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| The player does not move                    | The events are on the wrong object, or <code>obj_player</code> is not in the room       | Open <code>obj_player</code> and check the three Keyboard events; open the room and check the player is placed |
| The player keeps sliding after I let go     | The <b>No key</b> event is missing                                                      | Add <b>Keyboard: No key</b> with <i>*Stop movement*</i>                                                        |
| No stars appear                             | <code>obj_spawner</code> is not in the room, or its Create event does not set the alarm | Place the spawner; check <i>*Set Alarm 0 to 60*</i> is in its <b>Create</b> event                              |
| Only one star ever appears                  | The alarm is not set again at the end of <b>Alarm 0</b>                                 | Add <i>*Set Alarm 0 to 60*</i> as the last action of <b>Alarm 0</b>                                            |
| The score does not show                     | <code>obj_game_controller</code> is not in the room, or it has no Draw event            | Place it; add <i>*Draw score*</i> in a <b>Draw</b> event                                                       |
| The score never goes up                     | The Collision event is on the wrong object or has no <i>*Add to score*</i>              | Put <b>Collision with obj_star</b> on <code>obj_player</code>                                                  |
| My player disappears when it touches a star | I used <i>*Destroy this instance*</i> instead of <i>*Destroy other instance*</i>        | Use <b>Destroy other instance</b>                                                                              |

## Challenges

- **Try this (5 minutes):** change the star's fall speed, or the number of points for a star.
- **Push further:** make stars fall faster as the score goes up, or add a second kind of star worth more points.
- **Invent:** add a "bomb" object that ends the game if you catch it, or lives that you lose when a star reaches the bottom.

## Vocabulary

| Term      | What it means                                                            |
|-----------|--------------------------------------------------------------------------|
| Event     | Something that happens in the game (a key is pressed, two objects touch) |
| Action    | What an object does when an event happens                                |
| Alarm     | A timer; it counts down and then triggers its Alarm event                |
| Spawner   | An invisible object that creates other objects                           |
| Collision | When two objects touch                                                   |

## Check Yourself

Write your answers in the notes below.

- 1 What does an **alarm** do, and why does the spawner set it again at the end?
- 2 What is the difference between *\*Destroy this instance\** and *\*Destroy other instance\** in the collision event?
- 3 Which object draws the score, and why is it not the player?

## My Notes

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